

Human review packet: GJ18InformativeRatingSystems

Generated from byte-pinned source excerpts, the expanded PaperInterface specifications, and hash-bound raw-source-to-Spec screening records.

Paper: GJ18InformativeRatingSystems

Paper id: GJ18InformativeRatingSystems

Source version: GJ18 byte-pinned source and author-confirmed clarification inventory

Source record: audit/paper_statement_map.json

Rows: 18

Generated: 2026-08-18

Source URL: [official source](#)

Review status: 0/18 source-claim screens; 0/43 library prerequisite screens. This is a review worksheet while those direct checks remain pending; it is not a final validation receipt.
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How to use this packet

For each source claim, compare the exact source excerpts with the expanded PaperInterface specification. Mark up this PDF or fill the blank reviewer box in the TeX source; return the annotated file for reconciliation into the paper-local human-review log.

Open the interactive dashboard

The interactive dashboard is an optional alternative to using this PDF; it presents the same review surface rather than an additional review requirement. After forking and cloning (or pulling) EconCSLib, run the following from the repository root. The cache command is needed only the first time in a new clone.

```
cd <your-EconCSLib-checkout>
lake exe cache get
python3 scripts/review_dashboard.py --paper GJ18InformativeRatingSystems --serve
```

Open <http://127.0.0.1:8765/> in a browser and keep that terminal running while reviewing. The dashboard records saved annotations in this paper's reviewer-owned review record.

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Material library prerequisites

EconCSLib.Probability.ExponentialRateCertificate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

$(\mathbb{N} \rightarrow \mathbb{R}) \rightarrow \mathbb{R} \rightarrow \text{Prop}$

Exact Lean library declaration

```
structure ExponentialRateCertificate (p :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (rate :  $\mathbb{R}$ ) where
  eventually_pos :  $\forall^f n \text{ in } \text{atTop}, 0 < p\ n$ 
  has_rate : HasExponentialRate p rate
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteRatingLDPModel

Source connection: no explicit byte-pinned paper source connection is registered.

Lean declaration type (metadata)

Type $u_2 \rightarrow (\text{Rating} : \text{Type } u_3) \rightarrow [\text{Fintype Rating}] \rightarrow [\text{DecidableEq Rating}] \rightarrow \text{Type } (\max u_2 u_3)$

Exact Lean library declaration

```
structure FiniteRatingLDPModel (θ Rating : Type*) [Fintype Rating]
  [DecidableEq Rating] where
  typeLaw : θ → PMF Rating
  score : Rating → ℝ
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteRatingLDPModel.score

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{θ : Type u_2} →  
  {Rating : Type u_3} →  
    [inst : Fintype Rating] →  
      [inst_1 : DecidableEq Rating] →  
        (self : EconCSLib.Probability.FiniteRatingLDPModel θ Rating) → (a : Rating) → self.2 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteRatingLDPModel.typeLaw

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{θ : Type u_2} →  
  {Rating : Type u_3} →  
    [inst : Fintype Rating] →  
      [inst_1 : DecidableEq Rating] → (self : EconCSLib.Probability.FiniteRatingLDPModel θ Rating) → (a : θ) → self.1 a
```

Exact Lean library declaration

library declaration is not registered or discoverable

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.finiteMGF

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
[inst : Fintype α] → (μ : PMF α) → (score : α → ℝ) → (z : ℝ) → ∑ a, (μ a).toReal * Real.exp (z * score a)
```

Exact Lean library declaration

```
def finiteMGF (μ : PMF α) (score : α → ℝ) (z : ℝ) : ℝ :=  
  ∑ a : α, (μ a).toReal * Real.exp (z * score a)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.finiteLogMGF

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
  [inst : Fintype α] → (μ : PMF α) → (score : α → ℝ) → (z : ℝ) → Real.log (EconCSLib.Probability.finiteMGF μ score z)
```

Exact Lean library declaration

```
def finiteLogMGF (μ : PMF α) (score : α → ℝ) (z : ℝ) : ℝ :=  
  Real.log (finiteMGF μ score z)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteRatingLDPModel.logMGF

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{θ : Type u_2} →  
{Rating : Type u_3} →  
  [inst : Fintype Rating] →  
    [inst_1 : DecidableEq Rating] →  
      (M : EconCSLib.Probability.FiniteRatingLDPModel θ Rating) →  
        (t : θ) → (z : ℝ) → EconCSLib.Probability.finiteLogMGF (M.typeLaw t) M.score z
```

Exact Lean library declaration

```
def logMGF (M : FiniteRatingLDPModel θ Rating) (t : θ) (z : ℝ) : ℝ :=  
  finiteLogMGF (M.typeLaw t) M.score z
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.finiteLegendreValue

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
[inst : Fintype α] → (μ : PMF α) → (score : α → ℝ) → (a z : ℝ) → z * a - EconCSLib.Probability.finiteLogMGF μ score z
```

Exact Lean library declaration

```
def finiteLegendreValue (μ : PMF α) (score : α → ℝ)  
  (a z : ℝ) : ℝ :=  
  z * a - finiteLogMGF μ score z
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.finiteRateFunctionTop

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u 1} →  
[inst : Fintype α] →  
  (μ : PMF α) →  
    (score : α → ℝ) → (a : ℝ) → sSup (Set.range fun z => ↑(EconCSLib.Probability.finiteLegendreValue μ score a z))
```

Exact Lean library declaration

```
def finiteRateFunctionTop (μ : PMF α) (score : α → ℝ)  
  (a : ℝ) : WithTop ℝ :=  
  sSup (Set.range fun z : ℝ =>  
    (finiteLegendreValue μ score a z : WithTop ℝ))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteRatingLDPModel.rateFunctionTop

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{θ : Type u_2} →
{Rating : Type u_3} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPModel θ Rating) →
(t : θ) → (a : ℝ) → EconCSLib.Probability.finiteRateFunctionTop (M.typeLaw t) M.score a
```

Exact Lean library declaration

```
def rateFunctionTop (M : FiniteRatingLDPModel θ Rating)
  (t : θ) (a : ℝ) : WithTop ℝ :=
  finiteRateFunctionTop (M.typeLaw t) M.score a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.withTopRealScale

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(c : ℝ) → (x : WithTop ℝ) → WithTop.map (fun r => c * r) x
```

Exact Lean library declaration

```
def withTopRealScale (c : ℝ) (x : WithTop ℝ) : WithTop ℝ :=  
  WithTop.map (fun r : ℝ => c * r) x
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteRatingLDPModel.pairwiseRateObjectiveTop

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{θ : Type u_2} →
{Rating : Type u_3} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPModel θ Rating) →
(sampleRate : θ → ℝ) →
(hi lo : θ) →
(a : ℝ) →
  EconCSLib.Probability.withTopRealScale (sampleRate hi) (M.rateFunctionTop hi a) +
  EconCSLib.Probability.withTopRealScale (sampleRate lo) (M.rateFunctionTop lo a)
```

Exact Lean library declaration

```
def pairwiseRateObjectiveTop (M : FiniteRatingLDPModel θ Rating)
  (sampleRate : θ → ℝ) (hi lo : θ) (a : ℝ) : WithTop ℝ :=
  withTopRealScale (sampleRate hi) (M.rateFunctionTop hi a) +
  withTopRealScale (sampleRate lo) (M.rateFunctionTop lo a)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteRatingLDPModel.pairwiseThresholdRateTop

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{θ : Type u_2} →
{Rating : Type u_3} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPModel θ Rating) →
(sampleRate : θ → ℝ) → (hi lo : θ) → sInf (Set.range fun a => M.pairwiseRateObjectiveTop sampleRate hi lo a)
```

Exact Lean library declaration

```
noncomputable def pairwiseThresholdRateTop (M : FiniteRatingLDPModel θ Rating)
  (sampleRate : θ → ℝ) (hi lo : θ) : WithTop ℝ :=
  sInf (Set.range fun a : ℝ =>
    M.pairwiseRateObjectiveTop sampleRate hi lo a)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.finiteRateFunction

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
  [inst : Fintype α] →  
    (μ : PMF α) →  
      (score : α → ℝ) → (a : ℝ) → sSup (Set.range fun z => EconCSLib.Probability.finiteLegendreValue μ score a z)
```

Exact Lean library declaration

```
def finiteRateFunction (μ : PMF α) (score : α → ℝ)  
  (a : ℝ) : ℝ :=  
  sSup (Set.range fun z : ℝ => finiteLegendreValue μ score a z)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.FiniteRatingLDPModel.rateFunction

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{θ : Type u_2} →
{Rating : Type u_3} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPModel θ Rating) →
(t : θ) → (a : ℝ) → EconCSLib.Probability.finiteRateFunction (M.typeLaw t) M.score a
```

Exact Lean library declaration

```
def rateFunction (M : FiniteRatingLDPModel θ Rating)
  (t : θ) (a : ℝ) : ℝ :=
  finiteRateFunction (M.typeLaw t) M.score a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.logDecay

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(p : \mathbb{N} \rightarrow \mathbb{R}) \rightarrow (n : \mathbb{N}) \rightarrow -(1/n)^{-1} * \text{Real.log } (p \ n)$

Exact Lean library declaration

```
def logDecay (p :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (n :  $\mathbb{N}$ ) :  $\mathbb{R}$  :=  
  -((n :  $\mathbb{R}$ )-1) * Real.log (p n)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.HasExponentialRate

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\forall (p : \mathbb{N} \rightarrow \mathbb{R}) (rate : \mathbb{R}), \text{Filter.Tendsto } (\text{EconCSLib.Probability.logDecay } p) \text{ Filter.atTop } (nhds \text{ rate})$

Exact Lean library declaration

```
def HasExponentialRate (p :  $\mathbb{N} \rightarrow \mathbb{R}$ ) (rate :  $\mathbb{R}$ ) : Prop :=  
  Tendsto (logDecay p) atTop (nhds rate)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.floorSampleCount

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} → (sampleRate : Seller → ℝ) → (θ : Seller) → (k : ℕ) → [↑k * sampleRate θ]₊
```

Exact Lean library declaration

```
def floorSampleCount {Seller : Type*} (sampleRate : Seller → ℝ)
  (θ : Seller) (k : ℕ) : ℕ :=
  Nat.floor ((k : ℝ) * sampleRate θ)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.finiteChainJointFloorRatingSample

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →  
  (Rating : Type u_1) →  
    (sampleRate : Fin n → ℝ) →  
      (k : ℕ) → (θ : Fin n) → Fin (EconCSLib.Probability.floorSampleCount sampleRate θ k) → Rating
```

Exact Lean library declaration

```
abbrev finiteChainJointFloorRatingSample  
  {n : ℕ} (Rating : Type*) (sampleRate : Fin n → ℝ) (k : ℕ) :=  
    (θ : Fin n) → Fin (floorSampleCount sampleRate θ k) → Rating
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.finiteIdScoreSum

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{\alpha : \text{Type } u_1\} \rightarrow \{\iota : \text{Type } u_2\} \rightarrow [\text{inst} : \text{Fintype } \iota] \rightarrow (\text{score} : \alpha \rightarrow \mathbb{R}) \rightarrow (\text{sample} : \iota \rightarrow \alpha) \rightarrow \sum i, \text{score } (\text{sample } i)$

Exact Lean library declaration

```
def finiteIdScoreSum {ι : Type*} [Fintype ι]
  (score : α → ℝ) (sample : ι → α) : ℝ :=
  ∑ i : ι, score (sample i)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.finiteChainJointFloorAverageScore

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →
  {Rating : Type u_1} →
  [inst : Fintype Rating] →
  [inst_1 : DecidableEq Rating] →
  (M : EconCSLib.Probability.FiniteRatingLDPMModel (Fin n) Rating) →
  (sampleRate : Fin n → ℝ) →
  (k : ℕ) →
  (sample : EconCSLib.Probability.finiteChainJointFloorRatingSample Rating sampleRate k) →
  (θ : Fin n) →
  (↑(EconCSLib.Probability.floorSampleCount sampleRate θ k))-1 *
  EconCSLib.Probability.finiteIdScoreSum M.score (sample θ)
```

Exact Lean library declaration

```
def finiteChainJointFloorAverageScore
  {n : ℕ} {Rating : Type*} [Fintype Rating] [DecidableEq Rating]
  (M : FiniteRatingLDPMModel (Fin n) Rating) (sampleRate : Fin n → ℝ)
  (k : ℕ) (sample : finiteChainJointFloorRatingSample Rating sampleRate k)
  (θ : Fin n) : ℝ :=
  ((floorSampleCount sampleRate θ k : ℕ) : ℝ)-1 *
  finiteIdScoreSum M.score (sample θ)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfPi

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{ι : Type u_1} →  
  [inst : Fintype ι] →  
    [inst_1 : DecidableEq ι] →  
      {α : ι → Type u_2} →  
        [inst_2 : (i : ι) → Fintype (α i)] → (μ : (i : ι) → PMF (α i)) → PMF.ofFintype (fun f => ∏ i, (μ i) (f i)) ...
```

Exact Lean library declaration

```
noncomputable def pmfPi {ι : Type*} [Fintype ι] [DecidableEq ι]  
  {α : ι → Type*} [∀ i : ι, Fintype (α i)]  
  (μ : ∀ i : ι, PMF (α i)) : PMF ((i : ι) → α i) :=  
  PMF.ofFintype (fun f : (i : ι) → α i => ∏ i : ι, μ i (f i)) (by  
    classical  
    have hcoord : ∀ i : ι, ∑ a : α i, μ i a = 1 := by  
      intro i  
      rw [← PMF.tsum_coe (μ i), tsum_fintype]  
    calc  
      ∑ f : ((i : ι) → α i), ∏ i : ι, μ i (f i)  
      = ∑ f ∈ Fintype.piFinset  
        (fun i : ι => (Finset.univ : Finset (α i))),  
        ∏ i : ι, μ i (f i) := by  
        simp  
      _ = ∏ i : ι, ∑ a ∈ (Finset.univ : Finset (α i)), μ i a := by  
        symm  
        exact Finset.prod_univ_sum  
          (t := fun i : ι => (Finset.univ : Finset (α i)))  
          (f := fun i a => μ i a)  
      _ = 1 := by simp [hcoord])
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfProduct

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(ι : Type u_1) →
(α : Type u_2) →
  [inst : Fintype ι] →
    [inst_1 : DecidableEq ι] → [inst_2 : Fintype α] → (μ : PMF α) → PMF.ofFintype (fun f => [] i, μ (f i)) ...
```

Exact Lean library declaration

```
noncomputable def pmfProduct (ι α : Type*) [Fintype ι] [DecidableEq ι] [Fintype α]
  (μ : PMF α) : PMF (ι → α) :=
  PMF.ofFintype (fun f : ι → α => [] i : ι, μ (f i)) (by
    classical
    have hcoord : ∑ a : α, μ a = 1 := by
      rw [← PMF.tsum_coe μ, tsum_fintype]
    calc
      ∑ f : ι → α, [] i : ι, μ (f i)
      = [] i : ι, ∑ a : α, μ a := by
        symm
        simp using
          (Finset.prod_univ_sum
            (t := fun _i : ι => (Finset.univ : Finset α))
            (f := fun _i a => μ a))
      _ = 1 := by simp [hcoord])
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.finiteChainJointFloorRatingLaw

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{n : ℕ} →
  {Rating : Type u_1} →
  [inst : Fintype Rating] →
  [inst_1 : DecidableEq Rating] →
  (M : EconCSLib.Probability.FiniteRatingLDPMModel (Fin n) Rating) →
  (sampleRate : Fin n → ℝ) →
  (k : ℕ) →
    EconCSLib.pmfPi fun θ =>
      EconCSLib.pmfProduct (Fin (EconCSLib.Probability.floorSampleCount sampleRate θ k)) Rating (M.typeLaw θ)
```

Exact Lean library declaration

```
noncomputable def finiteChainJointFloorRatingLaw
  {n : ℕ} {Rating : Type*} [Fintype Rating] [DecidableEq Rating]
  (M : FiniteRatingLDPMModel (Fin n) Rating) (sampleRate : Fin n → ℝ)
  (k : ℕ) : PMF (finiteChainJointFloorRatingSample Rating sampleRate k) :=
  EconCSLib.pmfPi (fun θ : Fin n =>
    EconCSLib.pmfProduct (Fin (floorSampleCount sampleRate θ k)) Rating
      (M.typeLaw θ))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfProd

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
{β : Type u_2} →  
  [inst : Fintype α] → [inst_1 : Fintype β] → (μ : PMF α) → (ν : PMF β) → PMF.ofFintype (fun p => μ p.1 * ν p.2) ...
```

Exact Lean library declaration

```
noncomputable def pmfProd {α β : Type*} [Fintype α] [Fintype β]  
  (μ : PMF α) (ν : PMF β) : PMF (α × β) :=  
  PMF.ofFintype (fun p : α × β => μ p.1 * ν p.2) (by  
    classical  
    have hμ : ∑ a : α, μ a = 1 := by  
      rw [← PMF.tsum_coe μ, tsum_fintype]  
    have hν : ∑ b : β, ν b = 1 := by  
      rw [← PMF.tsum_coe ν, tsum_fintype]  
    calc  
      ∑ p : α × β, μ p.1 * ν p.2  
      = ∑ a : α, ∑ b : β, μ a * ν b := by  
        rw [Fintype.sum_prod_type]  
      = ∑ a : α, μ a * (∑ b : β, ν b) := by  
        simp [Finset.mul_sum]  
      = 1 := by  
        simp [hμ, hν])
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoSampleRatingLaw

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPModel Seller Rating) →
(hi lo : Seller) →
(nHi nLo : ℕ) →
  EconCSLib.pmfProd (EconCSLib.pmfProduct (Fin nHi) Rating (M.typeLaw hi))
    (EconCSLib.pmfProduct (Fin nLo) Rating (M.typeLaw lo))
```

Exact Lean library declaration

```
noncomputable def twoSampleRatingLaw
  {Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]
  (M : FiniteRatingLDPModel Seller Rating) (hi lo : Seller)
  (nHi nLo : ℕ) :
  PMF ((Fin nHi → Rating) × (Fin nLo → Rating)) :=
  EconCSLib.pmfProd
    (EconCSLib.pmfProduct (Fin nHi) Rating (M.typeLaw hi))
    (EconCSLib.pmfProduct (Fin nLo) Rating (M.typeLaw lo))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoSampleScoreGapSum

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPMModel Seller Rating) →
{nHi nLo : ℕ} →
(cHi cLo : ℝ) →
(sample : (Fin nHi → Rating) × (Fin nLo → Rating)) →
cHi * EconCSLib.Probability.finiteIidScoreSum M.score sample.1 -
cLo * EconCSLib.Probability.finiteIidScoreSum M.score sample.2
```

Exact Lean library declaration

```
def twoSampleScoreGapSum
{Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]
(M : FiniteRatingLDPMModel Seller Rating) {nHi nLo : ℕ}
(cHi cLo : ℝ)
(sample : (Fin nHi → Rating) × (Fin nLo → Rating)) : ℝ :=
cHi * finiteIidScoreSum M.score sample.1 -
cLo * finiteIidScoreSum M.score sample.2
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfExp

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$\{\alpha : \text{Type } u_1\} \rightarrow [\text{inst} : \text{Fintype } \alpha] \rightarrow [\text{DecidableEq } \alpha] \rightarrow (\mu : \text{PMF } \alpha) \rightarrow (f : \alpha \rightarrow \mathbb{R}) \rightarrow \sum a, (\mu a).toReal * f a$

Exact Lean library declaration

```
noncomputable def pmfExp {α : Type*} [Fintype α] [DecidableEq α]
  (μ : PMF α) (f : α → ℝ) : ℝ :=
  ∑ a, (μ a).toReal * f a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.pmfProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{α : Type u_1} →  
  [inst : Fintype α] →  
    [inst_1 : DecidableEq α] →  
      (μ : PMF α) → (p : α → Prop) → [inst_2 : DecidablePred p] → EconCSLib.pmfExp μ fun a => if p a then 1 else 0
```

Exact Lean library declaration

```
noncomputable def pmfProb {α : Type*} [Fintype α] [DecidableEq α]  
  (μ : PMF α) (p : α → Prop) [DecidablePred p] : ℝ :=  
  pmfExp μ (fun a => if p a then 1 else 0)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoSampleScoreGapStrictLeftProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPMModel Seller Rating) →
(hi lo : Seller) →
(nHi nLo : ℕ) →
(cHi cLo : ℝ) →
EconCSLib.pmfProb (EconCSLib.Probability.twoSampleRatingLaw M hi lo nHi nLo) fun sample =>
EconCSLib.Probability.twoSampleScoreGapSum M cHi cLo sample < 0
```

Exact Lean library declaration

```
def twoSampleScoreGapStrictLeftProb
{Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]
(M : FiniteRatingLDPMModel Seller Rating) (hi lo : Seller)
(nHi nLo : ℕ) (cHi cLo : ℝ) : ℝ :=
EconCSLib.pmfProb (twoSampleRatingLaw M hi lo nHi nLo)
(fun sample => twoSampleScoreGapSum M cHi cLo sample < 0)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoSampleScoreGapStrictRightProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPModel Seller Rating) →
(hi lo : Seller) →
(nHi nLo : ℕ) →
(cHi cLo : ℝ) →
EconCSLib.pmfProb (EconCSLib.Probability.twoSampleRatingLaw M hi lo nHi nLo) fun sample =>
0 < EconCSLib.Probability.twoSampleScoreGapSum M cHi cLo sample
```

Exact Lean library declaration

```
def twoSampleScoreGapStrictRightProb
{Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]
(M : FiniteRatingLDPModel Seller Rating) (hi lo : Seller)
(nHi nLo : ℕ) (cHi cLo : ℝ) : ℝ :=
EconCSLib.pmfProb (twoSampleRatingLaw M hi lo nHi nLo)
(fun sample => 0 < twoSampleScoreGapSum M cHi cLo sample)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoSamplePkObjectiveProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPModel Seller Rating) →
(hi lo : Seller) →
(nHi nLo : ℕ) →
(cHi cLo : ℝ) →
  EconCSLib.Probability.twoSampleScoreGapStrictRightProb M hi lo nHi nLo cHi cLo -
  EconCSLib.Probability.twoSampleScoreGapStrictLeftProb M hi lo nHi nLo cHi cLo
```

Exact Lean library declaration

```
def twoSamplePkObjectiveProb
  {Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]
  (M : FiniteRatingLDPModel Seller Rating) (hi lo : Seller)
  (nHi nLo : ℕ) (cHi cLo : ℝ) : ℝ :=
  twoSampleScoreGapStrictRightProb M hi lo nHi nLo cHi cLo -
  twoSampleScoreGapStrictLeftProb M hi lo nHi nLo cHi cLo
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoSampleFloorPkObjectiveProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPModel Seller Rating) →
(sampleRate : Seller → ℝ) →
(hi lo : Seller) →
(k : ℕ) →
  have nHi := EconCSLib.Probability.floorSampleCount sampleRate hi k;
  have nLo := EconCSLib.Probability.floorSampleCount sampleRate lo k;
  EconCSLib.Probability.twoSamplePkObjectiveProb M hi lo nHi nLo (↑nHi)-1 (↑nLo)-1
```

Exact Lean library declaration

```
def twoSampleFloorPkObjectiveProb
  {Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]
  (M : FiniteRatingLDPModel Seller Rating) (sampleRate : Seller → ℝ)
  (hi lo : Seller) (k : ℕ) : ℝ :=
  let nHi := floorSampleCount sampleRate hi k
  let nLo := floorSampleCount sampleRate lo k
  twoSamplePkObjectiveProb M hi lo nHi nLo
  ((nHi : ℝ)-1) ((nLo : ℝ)-1)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.finiteFloorPkObjective

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
{Pair : Type u_3} →
  [inst : Fintype Rating] →
  [inst_1 : DecidableEq Rating] →
  [inst_2 : Fintype Pair] →
    (M : EconCSLib.Probability.FiniteRatingLDPMModel Seller Rating) →
    (sampleRate : Seller → ℝ) →
    (pairHi pairLo : Pair → Seller) →
    (weight : Pair → ℝ) →
    (k : ℕ) →
      ∑ p,
        weight p *
          EconCSLib.Probability.twoSampleFloorPkObjectiveProb M sampleRate (pairHi p) (pairLo p) k
```

Exact Lean library declaration

```
def finiteFloorPkObjective
  {Seller Rating Pair : Type*} [Fintype Rating] [DecidableEq Rating]
  [Fintype Pair]
  (M : FiniteRatingLDPMModel Seller Rating) (sampleRate : Seller → ℝ)
  (pairHi pairLo : Pair → Seller) (weight : Pair → ℝ) (k : ℕ) : ℝ :=
  ∑ p : Pair,
    weight p *
      twoSampleFloorPkObjectiveProb M sampleRate (pairHi p) (pairLo p) k
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.uniformPairWeight

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
(Pair : Type u_1) → [inst : Fintype Pair] → Pair → (↑(Fintype.card Pair))-1
```

Exact Lean library declaration

```
def uniformPairWeight (Pair : Type*) [Fintype Pair] : Pair → ℝ :=  
  fun _ => (Fintype.card Pair : ℝ)-1
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.finiteUniformFloorPkObjective

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
{Pair : Type u_3} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
[inst_2 : Fintype Pair] →
(M : EconCSLib.Probability.FiniteRatingLDPModel Seller Rating) →
(sampleRate : Seller → ℝ) →
(pairHi pairLo : Pair → Seller) →
(k : ℕ) →
EconCSLib.Probability.finiteFloorPkObjective M sampleRate pairHi pairLo
(EconCSLib.Probability.uniformPairWeight Pair) k
```

Exact Lean library declaration

```
def finiteUniformFloorPkObjective
{Seller Rating Pair : Type*} [Fintype Rating] [DecidableEq Rating]
[Fintype Pair]
(M : FiniteRatingLDPModel Seller Rating) (sampleRate : Seller → ℝ)
(pairHi pairLo : Pair → Seller) (k : ℕ) : ℝ :=
finiteFloorPkObjective M sampleRate pairHi pairLo (uniformPairWeight Pair) k
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.pairwiseSellerThresholdRateTop

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →  
  {Rating : Type u_2} →  
    [inst : Fintype Rating] →  
      [inst_1 : DecidableEq Rating] →  
        (M : EconCSLib.Probability.FiniteRatingLDPMModel Seller Rating) →  
          (sampleRate : Seller → ℝ) → (hi lo : Seller) → M.pairwiseThresholdRateTop sampleRate hi lo
```

Exact Lean library declaration

```
noncomputable def pairwiseSellerThresholdRateTop  
  {Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]  
  (M : FiniteRatingLDPMModel Seller Rating) (sampleRate : Seller → ℝ)  
  (hi lo : Seller) : WithTop ℝ :=  
  M.pairwiseThresholdRateTop sampleRate hi lo
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.ratingLogMGF

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →  
  {Rating : Type u_2} →  
    [inst : Fintype Rating] →  
      [inst_1 : DecidableEq Rating] →  
        (M : EconCSLib.Probability.FiniteRatingLDPMModel Seller Rating) → (θ : Seller) → (z : ℝ) → M.logMGF θ z
```

Exact Lean library declaration

```
def ratingLogMGF {Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]  
  (M : FiniteRatingLDPMModel Seller Rating) (θ : Seller) (z : ℝ) : ℝ :=  
  M.logMGF θ z
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.ratingRateFunction

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →  
  {Rating : Type u_2} →  
    [inst : Fintype Rating] →  
      [inst_1 : DecidableEq Rating] →  
        (M : EconCSLib.Probability.FiniteRatingLDPModel Seller Rating) → (θ : Seller) → (a : ℝ) → M.rateFunction θ a
```

Exact Lean library declaration

```
def ratingRateFunction {Seller Rating : Type*} [Fintype Rating]  
  [DecidableEq Rating]  
  (M : FiniteRatingLDPModel Seller Rating) (θ : Seller) (a : ℝ) : ℝ :=  
  M.rateFunction θ a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.ratingRateFunctionTop

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →  
  {Rating : Type u_2} →  
    [inst : Fintype Rating] →  
      [inst_1 : DecidableEq Rating] →  
        (M : EconCSLib.Probability.FiniteRatingLDPMModel Seller Rating) → (θ : Seller) → (a : ℝ) → M.rateFunctionTop θ a
```

Exact Lean library declaration

```
def ratingRateFunctionTop {Seller Rating : Type*} [Fintype Rating]  
  [DecidableEq Rating]  
  (M : FiniteRatingLDPMModel Seller Rating) (θ : Seller) (a : ℝ) :  
    WithTop ℝ :=  
  M.rateFunctionTop θ a
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoSampleScoreGapLeftTailProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPMModel Seller Rating) →
(hi lo : Seller) →
(nHi nLo : ℕ) →
(cHi cLo : ℝ) →
EconCSLib.pmfProb (EconCSLib.Probability.twoSampleRatingLaw M hi lo nHi nLo) fun sample =>
EconCSLib.Probability.twoSampleScoreGapSum M cHi cLo sample ≤ 0
```

Exact Lean library declaration

```
def twoSampleScoreGapLeftTailProb
{Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]
(M : FiniteRatingLDPMModel Seller Rating) (hi lo : Seller)
(nHi nLo : ℕ) (cHi cLo : ℝ) : ℝ :=
EconCSLib.pmfProb (twoSampleRatingLaw M hi lo nHi nLo)
(fun sample => twoSampleScoreGapSum M cHi cLo sample ≤ 0)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.Probability.twoSampleFloorScoreGapLeftTailProb

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

```
{Seller : Type u_1} →
{Rating : Type u_2} →
[inst : Fintype Rating] →
[inst_1 : DecidableEq Rating] →
(M : EconCSLib.Probability.FiniteRatingLDPModel Seller Rating) →
(sampleRate : Seller → ℝ) →
(hi lo : Seller) →
(k : ℕ) →
  have nHi := EconCSLib.Probability.floorSampleCount sampleRate hi k;
  have nLo := EconCSLib.Probability.floorSampleCount sampleRate lo k;
  EconCSLib.Probability.twoSampleScoreGapLeftTailProb M hi lo nHi nLo (↑nHi)-1 (↑nLo)-1
```

Exact Lean library declaration

```
def twoSampleFloorScoreGapLeftTailProb
  {Seller Rating : Type*} [Fintype Rating] [DecidableEq Rating]
  (M : FiniteRatingLDPModel Seller Rating) (sampleRate : Seller → ℝ)
  (hi lo : Seller) (k : ℕ) : ℝ :=
  let nHi := floorSampleCount sampleRate hi k
  let nLo := floorSampleCount sampleRate lo k
  twoSampleScoreGapLeftTailProb M hi lo nHi nLo
  ((nHi : ℝ)-1) ((nLo : ℝ)-1)
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

EconCSLib.uniformPMF

Source connection: no explicit byte-pinned paper source connection is registered.

Lean-expanded library semantic target

$(\alpha : \text{Type } u_1) \rightarrow [\text{inst} : \text{Fintype } \alpha] \rightarrow [\text{inst}_1 : \text{Nonempty } \alpha] \rightarrow \text{PMF.ofFintype } (\text{fun } x \Rightarrow (\uparrow(\text{Fintype.card } \alpha))^{-1}) \dots$

Exact Lean library declaration

```
noncomputable def uniformPMF (α : Type*) [Fintype α] [Nonempty α] : PMF α :=
  PMF.ofFintype (fun _ : α => ((Fintype.card α : ENNReal)⁻¹)) (by
    rw [Finset.sum_const, Finset.card_univ]
    have hcard : (Fintype.card α : ENNReal) ≠ 0 := by
      exact_mod_cast (Fintype.card_pos_iff.mpr <Nonempty α>.ne')
    simp [nsmul_eq_mul] using
      (ENNReal.mul_inv_cancel hcard (ENNReal.natCast_ne_top (Fintype.card α))))
```

Recorded source-to-library screening Verdict: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Source claims

Review row 1

Source locator: cited publication, lines 669-670

Verbatim source input

{\bf Rating accumulation}. Sellers accumulate ratings over time by matching with buyers. At each time step, each seller matches
→ with at most a single buyer. We make one key assumption that drives the accumulation of ratings: in particular, that sellers
→ of higher quality are more likely to be matched. %
We consider an analysis that is asymptotic in the number of ratings received by sellers and so we model this visibility benefit
→ by assuming that {\em sellers of higher quality accumulate ratings at a faster rate}. In particular, we assume the existence
→ of a nondecreasing {\em match function} $g(\theta)$, where a seller of type θ receives $n_k(\theta) = \lfloor k \cdot g(\theta) \rfloor$ matches, and thus ratings, up to time k .

Expanded PaperInterface specification

$\forall \{ \text{Seller} : \text{Type } u_1 \} \text{ (sampleRate : Seller} \rightarrow \mathbb{R}) \text{ (theta : Seller) (k : } \mathbb{N} \text{),}$
EconCSLib.Probability.floorSampleCount sampleRate theta k = $\lfloor k * \text{sampleRate theta} \rfloor$.

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_definition_floor_sample_count_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 2

Source locator: cited publication, lines 697-702

Verbatim source input

Let $y_0(\theta), y_1(\theta), y_2(\theta), \dots$ be the sequence of ratings received by the seller of type θ . The $\{em \rightarrow \text{aggregate score}\}$ up to time k of this seller is the average score from ratings received:

$$x_k(\theta) = \frac{1}{n_k(\theta)} \sum_{\ell = 0}^{n_k(\theta)} \phi(y_\ell(\theta)).$$

(We presume $x_0(\theta) = 0$ for all θ .) Since $\phi(y) \in [0,1]$ for all y , the score x_k also lies in $[0,1]$.

Approved corrected review target The archival source above is not asserted equivalent to this replacement.

For each seller type θ and horizon k , if $n_k(\theta) = \text{floor}(k \cdot g(\theta))$, the aggregate score averages exactly $n_k(\theta)$ rating scores over a finite family indexed by $\text{Fin } n_k(\theta)$; the zero-sample convention remains explicit.

Archival source anchor: cited publication, lines 697-702

Recorded basis Author-approved corrected finite-chain model recorded in docs/GOVERNING_CORRECTED_MODEL_2026-07-24.md.

Expanded PaperInterface specification

```
∀ {n : ℕ} {Rating : Type u_1} [inst : Fintype Rating] [inst_1 : DecidableEq Rating]
  (M : EconCSLib.Probability.FiniteRatingLDPMModel (Fin n) Rating) (sampleRate : Fin n → ℝ) (k : ℕ)
  (sample : EconCSLib.Probability.finiteChainJointFloorRatingSample Rating sampleRate k) (theta : Fin n),
  EconCSLib.Probability.finiteChainJointFloorAverageScore M sampleRate k sample theta =
    (↑(EconCSLib.Probability.floorSampleCount sampleRate theta k))⁻¹ * ∑ i, M.score (sample theta i)
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_definition_aggregate_score_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 3

Source locator: cited publication, lines 721-725

Verbatim source input

Formally, given $\theta_1 > \theta_2$, define $P_k(\theta_1, \theta_2)$ as follows:

```
\begin{equation}
\label{eq:Pk}
P_k(\theta_1, \theta_2) = \mu_k(x_k(\theta_1) > x_k(\theta_2) | \theta_1, \theta_2) - \mu_k(x_k(\theta_1) < x_k(\theta_2) | \theta_1,
\leftrightarrow \theta_2).
\end{equation}
```

Expanded PaperInterface specification

```
∀ {n : ℕ} {Omega : Type u_1} [inst : Fintype Omega] [inst_1 : DecidableEq Omega] (mu : PMF Omega)
(score : Omega → Fin n → ℝ) (p : GJ18InformativeRatingSystems.finiteChainOrderedPair n),
GJ18InformativeRatingSystems.sourceStatePairwisePk mu score p =
(EconCSLib.pmfProb mu fun omega =>
  score omega (GJ18InformativeRatingSystems.finiteChainOrderedPairHi p) >
  score omega (GJ18InformativeRatingSystems.finiteChainOrderedPairLo p)) -
EconCSLib.pmfProb mu fun omega =>
  score omega (GJ18InformativeRatingSystems.finiteChainOrderedPairHi p) <
  score omega (GJ18InformativeRatingSystems.finiteChainOrderedPairLo p)
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_definition_pairwise_objective_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 4

Source locator: cited publication, lines 728-733

Verbatim source input

We consider the problem of maximizing the following objective, a scaled version of {\em Kendall's τ rank correlation} $\hookrightarrow$ between the estimated ranking of sellers and the true ranking:

$$\begin{aligned} &\backslash\mathrm{begin}\{equation\} \\ &\backslash\mathrm{label}\{eq:objective\} \\ &W_k = \frac{2}{M(M-1)} \sum_{\theta_1 > \theta_2 \in \Theta} P_k(\theta_1, \theta_2) \\ &\backslash\mathrm{end}\{equation\} \end{aligned}$$

The coefficient ensures that $\{W\}_k$ remains bounded even as M increases. This objective depends on the model primitives R $\hookrightarrow$ (rater behavior) and g (matching rates), as well as the platform's decisions Y (levels) and ϕ (score).

Expanded PaperInterface specification

$\forall \{n : \mathbb{N}\} \{ \Omega : \text{Type } u_1 \} [inst : \text{Fintype } \Omega] [inst_1 : \text{DecidableEq } \Omega] (\mu : \text{PMF } \Omega)$
(score : $\Omega \rightarrow \text{Fin } n \rightarrow \mathbb{R}$),
GJ18InformativeRatingSystems.sourceStateWk μ score =
 $2 / (n * (n - 1)) * \sum p, \text{GJ18InformativeRatingSystems.sourceStatePairwisePk } \mu \text{ score } p$

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_definition_uniform_ranking_objective_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 5

Source locator: cited publication, lines 729-733

Verbatim source input

$$W_k = \frac{2}{M(M-1)} \sum_{\theta_1 > \theta_2 \in \Theta} P_k(\theta_1, \theta_2)$$

The coefficient ensures that $\{W\}_k$ remains bounded even as M increases. This objective depends on the model primitives R (rater behavior) and g (matching rates), as well as the platform's decisions Y (levels) and ϕ (score).

Expanded PaperInterface specification

$\forall \{n : \mathbb{N}\} \{\Omega : \text{Type } u_1\} [\text{inst} : \text{Fintype } \Omega] [\text{inst}_1 : \text{DecidableEq } \Omega] (\mu : \text{PMF } \Omega)$
 $(\text{score} : \Omega \rightarrow \text{Fin } n \rightarrow \mathbb{R}),$
 $2 \leq n \rightarrow$
 $\text{GJ18InformativeRatingSystems.sourceStateWk } \mu \text{ score} = \text{GJ18InformativeRatingSystems.sourceStateUniformWk } \mu \text{ score}$

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_definition_uniform_ranking_normalization

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 6

Source locator: cited publication, lines 683

Verbatim source input

At each rating opportunity (i.e., match made), the seller receives a rating from the set $\mathcal{Y}$, and we assume that this rating depends only on the true quality of the seller. In particular, we presume that, given scale $\mathcal{Y}$, the probability a seller of type θ receives a rating y is $\rho(\theta, y | \mathcal{Y})$, with corresponding cumulative mass function $R(\theta, y | \mathcal{Y})$ reflecting the probability a seller of type θ receives a rating y or higher. In other words, the scale $\mathcal{Y}$ induces a joint distribution between the underlying seller quality and the rating choices buyers make. We make the natural assumptions that $R(\theta, y | \mathcal{Y})$ is strictly increasing in θ and strictly decreasing with y .

[Next verbatim source excerpt]

System state. We represent the state of the system defined above by a joint distribution $\mu_k(\theta, X)$, which gives the mass of sellers of type θ in Θ with aggregate score $x_k(\theta)$ in X at time k . Throughout our model presentation, we describe the system model as one emerging from interactions between individual buyers and sellers. However, we assume a unit mass of sellers (and some mass of buyers), and so all such descriptions should be viewed as illuminating the evolution of a joint distribution $\mu_k(\theta, X)$ of the types of sellers on the platform and their current scores. To formally describe the evolution of μ_k , let $E_k = \{\theta : n_k(\theta) = n_{k-1}(\theta) + 1\}$. These are the sellers who receive an additional rating at time k ; for all $\theta \in E_k^c$, $n_k(\theta) = n_{k-1}(\theta)$. Next, for each $x, x' \in [0, 1]$, define $\chi_k(x, x', \theta | \mathcal{Y}, \phi)$ as:

$$\chi_k(x, x', \theta | \mathcal{Y}, \phi) = \{y : n_k(\theta)x - n_{k-1}(\theta)x' = \phi(y)\}.$$

The set χ_k describes the rating(s) a seller of type θ at time k with aggregate score x' can receive to transition to aggregate score x . We then have:

$$\mu_{k+1}(\theta, X) = \int_{E_k} \int_{\Theta} \sum_y \chi_k(\theta, x, x' | \mathcal{Y}, \phi) \rho(\theta, y | \mathcal{Y}) dx \\ \mu_k(d\theta, dx) + \int_{E_k^c} \int_{\Theta} \mu_k(d\theta, dx).$$

It is straightforward but tedious to check that the preceding dynamics are well defined, given our primitives.

Platform objective. We assume that the platform wants the ranking of sellers by observed aggregate score to reflect the underlying true quality ranking as closely as possible.

Formally, given $\theta_1 > \theta_2$, define $P_k(\theta_1, \theta_2)$ as follows:

$$\begin{aligned} & \text{\texttt{\begin{equation}}} \\ & \text{\texttt{\label{eq:Pk}}} \\ & P_k(\theta_1, \theta_2) = \mu_k(x_k(\theta_1) > x_k(\theta_2) | \theta_1, \theta_2) - \mu_k(x_k(\theta_1) < x_k(\theta_2) | \theta_1, \\ & \quad \theta_2). \\ & \text{\texttt{\end{equation}}}} \end{aligned}$$

Expanded PaperInterface specification

```
∀ {n : ℕ} {Rating : Type u_1} [inst : Fintype Rating] [inst_1 : DecidableEq Rating]
  [DecidableEq (GJ18InformativeRatingSystems.finiteChainOrderedPair n)]
  (M : EconCSLib.Probability.FiniteRatingLDPModel (Fin n) Rating) (sampleRate : Fin n → ℝ)
  (p : GJ18InformativeRatingSystems.finiteChainOrderedPair n) (k : ℕ),
  GJ18InformativeRatingSystems.sourceStatePairwisePk
    (GJ18InformativeRatingSystems.iidFloorSourceStateLaw M sampleRate k)
    (GJ18InformativeRatingSystems.iidFloorSourceStateScore M sampleRate k) p =
  EconCSLib.Probability.twoSampleFloorPkObjectiveProb M sampleRate
    (GJ18InformativeRatingSystems.finiteChainOrderedPairHi p)
    (GJ18InformativeRatingSystems.finiteChainOrderedPairLo p) k
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_iid_completion_pairwise_objective_bridge

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 7

Source locator: cited publication, lines 683

Verbatim source input

At each rating opportunity (i.e., match made), the seller receives a rating from the set $\mathcal{Y}$, and we assume that this rating $\hookrightarrow$ depends only on the true quality of the seller. In particular, we presume that, given scale $\mathcal{Y}$, the probability a seller of type θ receives a rating y is $\rho(\theta, y | \mathcal{Y})$, with corresponding cumulative mass function $R(\theta, y | \mathcal{Y})$ $\hookrightarrow$ reflecting the probability a seller of type θ receives a rating y or higher. In other words, the scale $\mathcal{Y}$ induces a $\hookrightarrow$ joint distribution between the underlying seller quality and the rating choices buyers make. We make the natural assumptions $\hookrightarrow$ that $R(\theta, y | \mathcal{Y})$ is strictly increasing in θ and strictly decreasing with y .

[Next verbatim source excerpt]

Formally, given $\theta_1 > \theta_2$, define $P_k(\theta_1, \theta_2)$ as follows:

```
\begin{equation}
\label{eq:Pk}
P_k(\theta_1, \theta_2) = \mu_k(x_k(\theta_1) > x_k(\theta_2) | \theta_1, \theta_2) - \mu_k(x_k(\theta_1) < x_k(\theta_2) | \theta_1,
\hookrightarrow \theta_2).
\end{equation}
```

[Next verbatim source excerpt]

```
\begin{lemma}
  \lim_{k \rightarrow \infty} \frac{1}{k} \log \left[ \mu((x_k(\theta_1) - x_k(\theta_2)) \leq 0 | \theta_1,
  \hookrightarrow \theta_2) \right] = \inf_{a \in \mathcal{R}} \left\{ g(\theta_1) I(a | \theta_1) + g(\theta_2) I(a | \theta_2) \right\}
  \hookrightarrow \text{where } I(a | \theta) = \sup_z \{ z a - \Lambda(z | \theta) \}, \text{ } \Lambda(z | \theta) \text{ is the log moment generating}
  \hookrightarrow \text{function of a single sample from } x(\theta), \text{ and } g(\theta) \text{ is the sampling rate.} \label{lem:problessthan}
\end{lemma}
%
\proof
\indent \lim_{k \rightarrow \infty} \frac{1}{k} \log \left[ \mu((x_k(\theta_1) - x_k(\theta_2)) \leq 0 | \theta_1,
\hookrightarrow \theta_2) \right] =
\begin{align}
&= \lim_{k \rightarrow \infty} \frac{1}{k} \log \left[ \int_{a \in \mathcal{R}} \mu((x_k(\theta_1) = a | \theta_1) \right. \\
&\hookrightarrow \left. \mu(x_k(\theta_2) \geq a | \theta_2) da \right] \\
&= \lim_{k \rightarrow \infty} \frac{1}{k} \log \left[ \int_{a \in \mathcal{R}} e^{-kg(\theta_1) I(a | \theta_1)} \right. \\
&\hookrightarrow \left. e^{-kg(\theta_2) I(a | \theta_2)} da \right] \label{eqnpart:applyld} \\
&= \inf_{a \in \mathcal{R}} \left\{ g(\theta_1) I(a | \theta_1) + g(\theta_2) I(a | \theta_2) \right\} \text{ \& \textit{Laplace}} \\
&\hookrightarrow \text{principle}
\end{align}
%
```

Where $\text{eqref{eqnpart:applyld}}$ is a basic result from large deviations, and $kg(\theta_i)$ is the number of samples item of $\hookrightarrow$ quality θ_i has received. $\square$

Expanded PaperInterface specification

```
∀ {n : ℕ} {Rating : Type u_1} [inst : Fintype Rating] [inst_1 : DecidableEq Rating]
[DecidableEq (GJ18InformativeRatingSystems.finiteChainOrderedPair n)]
(M : EconCSLib.Probability.FiniteRatingLDPMModel (Fin n) Rating) (sampleRate : Fin n → ℝ)
(p : GJ18InformativeRatingSystems.finiteChainOrderedPair n) (k : ℕ),
GJ18InformativeRatingSystems.sourceStateScoreGapLeftTailProb
(GJ18InformativeRatingSystems.iidFloorSourceStateLaw M sampleRate k)
(GJ18InformativeRatingSystems.iidFloorSourceStateScore M sampleRate k) p =
EconCSLib.Probability.twoSampleFloorScoreGapLeftTailProb M sampleRate
(GJ18InformativeRatingSystems.finiteChainOrderedPairHi p)
(GJ18InformativeRatingSystems.finiteChainOrderedPairLo p) k
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_iid_completion_weak_inversion_bridge

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 8

Source locator: cited publication, lines 665

Verbatim source input

{\bf Sellers}. The system consists of a unit mass of sellers, each associated with a quality θ , which is (initially) unknown to the platform. We assume θ is drawn independently and uniformly at random from a finite and totally ordered set Θ , with $|\Theta| = M$. We use θ_i to denote the i th element of Θ within this order, for $0 \leq i < M$.

[Next verbatim source excerpt]

At each rating opportunity (i.e., match made), the seller receives a rating from the set $\mathcal{Y}$, and we assume that this rating depends only on the true quality of the seller. In particular, we presume that, given scale $\mathcal{Y}$, the probability a seller of type θ receives a rating y is $\rho(\theta, y | \mathcal{Y})$, with corresponding cumulative mass function $R(\theta, y | \mathcal{Y})$ reflecting the probability a seller of type θ receives a rating y or higher. In other words, the scale $\mathcal{Y}$ induces a joint distribution between the underlying seller quality and the rating choices buyers make. We make the natural assumptions that $R(\theta, y | \mathcal{Y})$ is strictly increasing in θ and strictly decreasing with y .

[Next verbatim source excerpt]

{\bf System state}. We represent the state of the system defined above by a joint distribution $\mu_k(\Theta, X)$, which gives the mass of sellers of type $\theta \in \Theta$ with aggregate score $x_k(\theta) \in X$ at time k . Throughout our model presentation, we describe the system model as one emerging from interactions between individual buyers and sellers. However, we assume a unit mass of sellers (and some mass of buyers), and so all such descriptions should be viewed as illuminating the evolution of a joint distribution $\mu_k(\Theta, X)$ of the types of sellers on the platform and their current scores. To formally describe the evolution of μ_k , let $E_k = \{\theta : n_k(\theta) = n_{k-1}(\theta) + 1\}$. These are the sellers who receive an additional rating at time k ; for all $\theta \in E_k^c$, $n_k(\theta) = n_{k-1}(\theta)$. Next, for each $x, x' \in [0, 1]$, define $\chi_k(x, x', \theta | \mathcal{Y}, \phi)$ as:

$$\chi_k(x, x', \theta | \mathcal{Y}, \phi) = \{y : n_k(\theta)x - n_{k-1}(\theta)x' = \phi(y)\}.$$

The set χ_k describes the rating(s) a seller of type θ at time k with aggregate score x can receive to transition to aggregate score x' . We then have:

$$\mu_{k+1}(\Theta, X) = \int_{E_k} \int_{\Theta^1} \int_X \sum_y \chi_k(\theta, x, x' | \mathcal{Y}, \phi) \rho(\theta, y | \mathcal{Y}) dx$$
$$+ \mu_k(d\theta, dx') + \int_{E_k^c} \int_X \mu_k(d\theta, dx').$$

It is straightforward but tedious to check that the preceding dynamics are well defined, given our primitives.

{\bf Platform objective}. We assume that the platform wants the ranking of sellers by observed aggregate score to reflect the underlying true quality ranking as closely as possible.

Formally, given $\theta_1 > \theta_2$, define $P_k(\theta_1, \theta_2)$ as follows:

$$P_k(\theta_1, \theta_2) = \mu_k(x_k(\theta_1) > x_k(\theta_2) | \theta_1, \theta_2) - \mu_k(x_k(\theta_1) < x_k(\theta_2) | \theta_1, \theta_2).$$

This expression captures the "errors" made by the ranking according to observed score. In particular, when $\theta_1 > \theta_2$ but $x_k(\theta_1) < x_k(\theta_2)$, the aggregate score ranking swaps the ordering of sellers θ_1 and θ_2 . Thus, a good rating system has large $P_k(\theta_1, \theta_2)$.

We consider the problem of maximizing the following objective, a scaled version of Kendall's τ rank correlation between the estimated ranking of sellers and the true ranking:

$$W_k = \frac{1}{M(M-1)} \sum_{\theta_1 > \theta_2 \in \Theta} P_k(\theta_1, \theta_2)$$

The coefficient ensures that W_k remains bounded even as M increases. This objective depends on the model primitives R (rater behavior) and g (matching rates), as well as the platform's decisions $\mathcal{Y}$ (levels) and ϕ (score).

Expanded PaperInterface specification

```
∀ {n : ℕ} {Rating : Type u_1} [inst : Fintype Rating] [inst_1 : DecidableEq Rating]
[DecidableEq (GJ18InformativeRatingSystems.finiteChainOrderedPair n)]
(M : EconCSLib.Probability.FiniteRatingLDPMModel (Fin n) Rating) (sampleRate : Fin n → ℝ) (k : ℕ),
2 ≤ n →
  GJ18InformativeRatingSystems.sourceStateWk (GJ18InformativeRatingSystems.iidFloorSourceStateLaw M sampleRate k)
  (GJ18InformativeRatingSystems.iidFloorSourceStateScore M sampleRate k) =
  EconCSLib.Probability.finiteUniformFloorPkObjective M sampleRate
  GJ18InformativeRatingSystems.finiteChainOrderedPairHi GJ18InformativeRatingSystems.finiteChainOrderedPairLo k
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_iid_completion_uniform_ranking_objective_bridge

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 9

Source locator: cited publication, lines 665

Verbatim source input

{\bf Sellers}. The system consists of a unit mass of sellers, each associated with a quality θ , which is
→ (initially) unknown to the platform. We assume θ is drawn independently and uniformly at random from a finite and
→ totally ordered set Θ , with $|\Theta| = M$. We use θ_i to denote the i th element of Θ within this
→ order, for $0 \leq i < M$.

Expanded PaperInterface specification

```
∀ {n m : ℕ} {M : EconCSLib.Probability.FiniteRatingLDPMModel (Fin n) (Fin (m + 1))} {sampleRate : Fin n → ℝ}
  (model : GJ18InformativeRatingSystems.ClarifiedSourceModel M sampleRate),
  model.type_prior = GJ18InformativeRatingSystems.clarifiedUniformTypePrior n ...
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.author_confirmed_clarified_uniform_type_prior

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 10

Source locator: cited publication, lines 683

Verbatim source input

At each rating opportunity (i.e., match made), the seller receives a rating from the set $\mathcal{Y}$, and we assume that this rating depends only on the true quality of the seller. In particular, we presume that, given scale $\mathcal{Y}$, the probability a seller of type θ receives a rating y is $\rho(\theta, y | \mathcal{Y})$, with corresponding cumulative mass function $R(\theta, y | \mathcal{Y})$ reflecting the probability a seller of type θ receives a rating y or higher. In other words, the scale $\mathcal{Y}$ induces a joint distribution between the underlying seller quality and the rating choices buyers make. We make the natural assumptions that $R(\theta, y | \mathcal{Y})$ is strictly increasing in θ and strictly decreasing with y .

[Next verbatim source excerpt]

{\bf System state}. We represent the state of the system defined above by a joint distribution $\mu_k(\Theta, X)$, which gives the mass of sellers of type $\theta \in \Theta$ with aggregate score $x_k(\theta) \in X$ at time k . Throughout our model presentation, we describe the system model as one emerging from interactions between individual buyers and sellers. However, we assume a unit mass of sellers (and some mass of buyers), and so all such descriptions should be viewed as illuminating the evolution of a joint distribution $\mu_k(\Theta, X)$ of the types of sellers on the platform and their current scores. To formally describe the evolution of μ_k , let $E_k = \{\theta : n_k(\theta) = n_{k-1}(\theta) + 1\}$. These are the sellers who receive an additional rating at time k ; for all $\theta \in E_k^c$, $n_k(\theta) = n_{k-1}(\theta)$. Next, for each $x, x' \in [0, 1]$, define $\chi_k(x, x', \theta | \mathcal{Y}, \phi)$ as:

$$\chi_k(x, x', \theta | \mathcal{Y}, \phi) = \{y : n_k(\theta)x - n_{k-1}(\theta)x' = \phi(y)\}.$$

The set χ_k describes the rating(s) a seller of type θ at time k with aggregate score x' can receive to transition to aggregate score x . We then have:

$$\mu_{k+1}(\Theta, X) = \int_{E_k} \int_0^1 \int_X \sum_y \chi_k(\theta, x, x' | \mathcal{Y}, \phi) \rho(\theta, y | \mathcal{Y}) dx \\ \mu_k(d\theta, dx') + \int_{E_k^c} \int_X \mu_k(d\theta, dx').$$

It is straightforward but tedious to check that the preceding dynamics are well defined, given our primitives.

Expanded PaperInterface specification

```
∀ {n m : ℕ} {M : EconCSLib.Probability.FiniteRatingLDPMModel (Fin n) (Fin (m + 1))} {sampleRate : Fin n → ℝ}
(model : GJ18InformativeRatingSystems.ClarifiedSourceModel M sampleRate) (k : ℕ),
model.stateLaw k = GJ18InformativeRatingSystems.iidFloorSourceStateLaw M sampleRate k
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.author_confirmed_clarified_iid_state_law

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 11

Source locator: cited publication, lines 754-761

Verbatim source input

```
\begin{restatable}{theorem}{lemhatW}
\label{lem:hatW}
\begin{equation}
\label{eq:rw}
r \triangleq - \lim_{k \rightarrow \infty} \frac{1}{k} \log (1 - \{W\}_k) = \min_{0 \leq i < M} \inf_{a \in \mathbb{R}} \left\{ \right.
\rightarrow g(\theta_{i+1}) I(a|\theta_{i+1}) + g(\theta_i) I(a|\theta_i) \left. \right\}
\end{equation}
where
 $I(a|\theta) = \sup_z \{ \int z a - \Lambda(z|\theta) \}$ , and  $\Lambda(z|\theta) = \log \sum_{y \in Y} \{\rho(\theta, y | Y)\}$ 
 $\rightarrow \exp(z \phi(y))$  is the log moment generating function of a single rating given to seller of type  $\theta$ .
```

[Next verbatim source excerpt]

```
\begin{lemma}
 $\lim_{k \rightarrow \infty} - \frac{1}{k} \log \left[ \mu((x_k(\theta_1) - x_k(\theta_2)) \leq 0 | \theta_1, \right.$ 
 $\rightarrow \theta_2) \left. \right] = \inf_{a \in \mathcal{R}} \left\{ g(\theta_1) I(a|\theta_1) + g(\theta_2) I(a|\theta_2) \right\}$ 
 $\rightarrow \}$  where  $I(a|\theta) = \sup_z \{ \int z a - \Lambda(z|\theta) \}$ ,  $\Lambda(z|\theta)$  is the log moment generating
 $\rightarrow$  function of a single sample from  $x(\theta_1)$ , and  $g(\theta)$  is the sampling rate.
\end{lemma}
```

[Next verbatim source excerpt]

```
\begin{restatable}{lemma}{lemPkld}
%
\label{lem:Pk_LD}
Given  $\theta_1, \theta_2$ , let  $\{P\}_k(\theta_1, \theta_2) = 1 - P_k(\theta_1, \theta_2)$ . Then:
\begin{equation}
\label{eq:LDrate}
- \lim_{k \rightarrow \infty} \frac{1}{k} \log \{P\}_k(\theta_1, \theta_2) = \inf_{a \in \mathcal{R}} \left\{ g(\theta_1) \right.
\rightarrow I(a|\theta_1) + g(\theta_2) I(a|\theta_2) \left. \right\},
\end{equation}
where  $I(a|\theta) = \sup_z \{ \int z a - \Lambda(z|\theta) \}$ , and  $\Lambda(z|\theta)$  is the log moment generating function
 $\rightarrow$  of a single rating given to seller of type  $\theta$ :
 $\Lambda(z|\theta) = \log \sum_{y \in Y} \rho(\theta, y | Y) \exp(z \phi(y))$ .
```

Expanded PaperInterface specification

```
∀ {Seller : Type u_1} {Rating : Type u_2} [inst : Fintype Rating] [inst_1 : DecidableEq Rating]
(M : EconCSLib.Probability.FiniteRatingLDPMModel Seller Rating) (theta : Seller) (z : ℝ),
GJ18InformativeRatingSystems.ProofBridge.sourcePaperLogMGF M theta z =
Real.log (∑ y, ((M.typeLaw theta) y).toReal * Real.exp (z * M.score y))
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_definition_log_mgf_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 12

Source locator: cited publication, lines 754-761

Verbatim source input

```
\begin{restatable}{theorem}{lemhatW}
\label{lem:hatW}
\begin{equation}
\label{eq:rw}
r \triangleq - \lim_{k \rightarrow \infty} \frac{1}{k} \log (1 - \{W\}_k) = \min_{0 \leq i < M} \inf_{a \in \mathbb{R}} \left\{ \right.
\rightarrow g(\theta_{i+1}) I(a|\theta_{i+1}) + g(\theta_i) I(a|\theta_i) \left. \right\}
\end{equation}
where
 $I(a|\theta) = \sup_z \{ z a - \Lambda(z|\theta) \}$ , and  $\Lambda(z|\theta) = \log \sum_{y \in Y} \{\rho(\theta, y | Y)\}$ 
 $\rightarrow \exp(z \phi(y))$  is the log moment generating function of a single rating given to seller of type  $\theta$ .
```

[Next verbatim source excerpt]

```
\begin{lemma}
 $\lim_{k \rightarrow \infty} - \frac{1}{k} \log \left[ \mu((x_k(\theta_1) - x_k(\theta_2)) \leq 0 | \theta_1, \right.$ 
 $\rightarrow \theta_2) \left. \right] = \inf_{a \in \mathcal{R}} \left\{ g(\theta_1) I(a|\theta_1) + g(\theta_2) I(a|\theta_2) \right\}$ 
 $\rightarrow \}$  where  $I(a|\theta) = \sup_z \{ z a - \Lambda(z|\theta) \}$ ,  $\Lambda(z|\theta)$  is the log moment generating
 $\rightarrow$  function of a single sample from  $x(\theta_1)$ , and  $g(\theta)$  is the sampling rate. \label{lem:problessthan}
\end{lemma}
```

[Next verbatim source excerpt]

```
\begin{restatable}{lemma}{lemPkld}
%
\label{lem:Pk_LD}
Given  $\theta_1, \theta_2$ , let  $\{P\}_k(\theta_1, \theta_2) = 1 - P_k(\theta_1, \theta_2)$ . Then:
\begin{equation}
\label{eq:LDrate}
- \lim_{k \rightarrow \infty} \frac{1}{k} \log \{P\}_k(\theta_1, \theta_2) = \inf_{a \in \mathcal{R}} \left\{ g(\theta_1) \right.
\rightarrow I(a|\theta_1) + g(\theta_2) I(a|\theta_2) \left. \right\},
\end{equation}
where  $I(a|\theta) = \sup_z \{ z a - \Lambda(z|\theta) \}$ , and  $\Lambda(z|\theta)$  is the log moment generating function
 $\rightarrow$  of a single rating given to seller of type  $\theta$ :
 $\Lambda(z|\theta) = \log \sum_{y \in Y} \rho(\theta, y | Y) \exp(z \phi(y))$ .
```

Expanded PaperInterface specification

```
∀ {Seller : Type u_1} {Rating : Type u_2} [inst : Fintype Rating] [inst_1 : DecidableEq Rating]
(M : EconCSLib.Probability.FiniteRatingLDPMModel Seller Rating) (theta : Seller) (a : ℝ),
GJ18InformativeRatingSystems.ProofBridge.sourcePaperRateFunction M theta a =
sSup (Set.range fun z => z * a - GJ18InformativeRatingSystems.ProofBridge.sourcePaperLogMGF M theta z)
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_definition_rate_function_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 13

Source locator: cited publication, lines 754-761

Verbatim source input

```
\begin{restatable}{theorem}{lemhatW}
\label{lem:hatW}
\begin{equation}
\label{eq:rw}
r \triangleq - \lim_{k \rightarrow \infty} \frac{1}{k} \log (1 - \{W\}_k) = \min_{0 \leq i < M} \inf_{a \in \mathbb{R}} \left\{ \right.
\rightarrow g(\theta_{i+1}) I(a|\theta_{i+1}) + g(\theta_i) I(a|\theta_i) \left. \right\}
\end{equation}
where
 $I(a|\theta) = \sup_z \{ \int z a - \Lambda(z|\theta) \}$ , and  $\Lambda(z|\theta) = \log \sum_{y \in Y} \{\rho(\theta, y | Y)\}$ 
 $\rightarrow \exp(z \phi(y))$  is the log moment generating function of a single rating given to seller of type  $\theta$ .
```

[Next verbatim source excerpt]

```
\begin{lemma}
 $\lim_{k \rightarrow \infty} - \frac{1}{k} \log \left[ \mu((x_k(\theta_1) - x_k(\theta_2)) \leq 0 | \theta_1, \right.$ 
 $\rightarrow \theta_2) \left. \right] = \inf_{a \in \mathcal{R}} \left\{ g(\theta_1) I(a|\theta_1) + g(\theta_2) I(a|\theta_2) \right\}$ 
 $\rightarrow \}$  where  $I(a|\theta) = \sup_z \{ \int z a - \Lambda(z|\theta) \}$ ,  $\Lambda(z|\theta)$  is the log moment generating
 $\rightarrow$  function of a single sample from  $x(\theta_1)$ , and  $g(\theta)$  is the sampling rate.
\end{lemma}
```

[Next verbatim source excerpt]

```
\begin{restatable}{lemma}{lemPkld}
%
\label{lem:Pk_LD}
Given  $\theta_1, \theta_2$ , let  $\{P\}_k(\theta_1, \theta_2) = 1 - P_k(\theta_1, \theta_2)$ . Then:
\begin{equation}
\label{eq:LDrate}
- \lim_{k \rightarrow \infty} \frac{1}{k} \log \{P\}_k(\theta_1, \theta_2) = \inf_{a \in \mathcal{R}} \left\{ g(\theta_1) \right.
\rightarrow I(a|\theta_1) + g(\theta_2) I(a|\theta_2) \left. \right\},
\end{equation}
where  $I(a|\theta) = \sup_z \{ \int z a - \Lambda(z|\theta) \}$ , and  $\Lambda(z|\theta)$  is the log moment generating function
 $\rightarrow$  of a single rating given to seller of type  $\theta$ :
 $\Lambda(z|\theta) = \log \sum_{y \in Y} \rho(\theta, y | Y) \exp(z \phi(y))$ .
```

Expanded PaperInterface specification

```
∀ {Seller : Type u_1} {Rating : Type u_2} [inst : Fintype Rating] [inst_1 : DecidableEq Rating]
(M : EconCSLib.Probability.FiniteRatingLDPMModel Seller Rating) (theta : Seller) (a : ℝ),
GJ18InformativeRatingSystems.ProofBridge.sourcePaperRateFunctionTop M theta a =
sSup (Set.range fun z => ↑z * ↑a - ↑(GJ18InformativeRatingSystems.ProofBridge.sourcePaperLogMGF M theta z))
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_definition_extended_rate_function_formula

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 14

Source locator: cited publication, lines 1633-1635

Verbatim source input

[No record available.]

Approved corrected review target The archival source above is not asserted equivalent to this replacement.

For every ordered high/low seller pair in the author-approved finite ordinal IID model, with positive match rates, a nontrivial
↪ rating scale, corrected ordinal tails, and an extended-real rate convention, the source-state weak-inversion probability has
↪ an exponential-rate certificate with a finite real representative of the support-safe pairwise threshold rate.

Archival source anchor: cited publication, lines 1633-1635

Recorded basis Author-approved corrected finite-chain model recorded in docs/GOVERNING_CORRECTED_MODEL_2026-07-24.md.

Expanded PaperInterface specification

```
∀ {n m : ℕ} [DecidableEq (GJ18InformativeRatingSystems.finiteChainOrderedPair n)]
(M : EconCSLib.Probability.FiniteRatingLDPModel (Fin n) (Fin (m + 1))) (sampleRate : Fin n → ℝ),
GJ18InformativeRatingSystems.FiniteOrdinalSourceModel M sampleRate →
(∀ (theta : Fin n), 0 < sampleRate theta) →
  1 ≤ m →
    ∀ (p : GJ18InformativeRatingSystems.finiteChainOrderedPair n),
      ∃ rate,
        EconCSLib.Probability.pairwiseSellerThresholdRateTop M sampleRate
          (GJ18InformativeRatingSystems.finiteChainOrderedPairHi p)
          (GJ18InformativeRatingSystems.finiteChainOrderedPairLo p) =
          ↑rate ∧
          EconCSLib.Probability.ExponentialRateCertificate
            (fun k =>
              GJ18InformativeRatingSystems.sourceStateScoreGapLeftTailProb
                (GJ18InformativeRatingSystems.iidFloorSourceStateLaw M sampleRate k)
                (GJ18InformativeRatingSystems.iidFloorSourceStateScore M sampleRate k) p)
            rate
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_appendix_problessthan_score_gap_rate_iid_completion

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 15

Source locator: cited publication, lines 1670-1681

Verbatim source input

[No record available.]

Approved corrected review target The archival source above is not asserted equivalent to this replacement.

For every ordered high/low seller pair in the author-approved finite ordinal IID model, with positive match rates, a nontrivial
↪ rating scale, corrected ordinal tails, and an extended-real rate convention, one minus the source-state P_k objective has an
↪ exponential-rate certificate with the same finite real representative of the support-safe pairwise threshold rate.

Archival source anchor: cited publication, lines 1670-1681

Recorded basis Author-approved corrected finite-chain model recorded in docs/GOVERNING_CORRECTED_MODEL_2026-07-24.md.

Expanded PaperInterface specification

```
∀ {n m : ℕ} [DecidableEq (GJ18InformativeRatingSystems.finiteChainOrderedPair n)]
(M : EconCSLib.Probability.FiniteRatingLDPModel (Fin n) (Fin (m + 1))) (sampleRate : Fin n → ℝ),
GJ18InformativeRatingSystems.FiniteOrdinalSourceModel M sampleRate →
(∀ (theta : Fin n), 0 < sampleRate theta) →
  1 ≤ m →
    ∀ (p : GJ18InformativeRatingSystems.finiteChainOrderedPair n),
      ∃ rate,
        EconCSLib.Probability.pairwiseSellerThresholdRateTop M sampleRate
          (GJ18InformativeRatingSystems.finiteChainOrderedPairHi p)
          (GJ18InformativeRatingSystems.finiteChainOrderedPairLo p) =
            ↑rate ∧
            EconCSLib.Probability.ExponentialRateCertificate
              (fun k =>
                1 -
                  GJ18InformativeRatingSystems.sourceStatePairwisePk
                    (GJ18InformativeRatingSystems.iidFloorSourceStateLaw M sampleRate k)
                    (GJ18InformativeRatingSystems.iidFloorSourceStateScore M sampleRate k) p)
              rate
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_appendix_pk_ld_complement_rate_iid_completion

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 16

Source locator: cited publication, lines 744-745

Verbatim source input

No single choice of Y and ϕ can simultaneously optimize W_k for all k : some designs may be effective in separating the best sellers from the worst quickly, but then never separate all sellers. Further, as long as $\phi(y)$ is strictly increasing, then because $R(\theta, y)$ is strictly increasing in θ , we have, for all $\theta_1 \neq \theta_2$, and all choices of Y and ϕ :

$$\lim_{k \rightarrow \infty} P_k(\theta_1, \theta_2) = 1.$$

Using the bounded convergence theorem we conclude that $\lim_{k \rightarrow \infty} W_k = 1$, independent of the design choice Y and ϕ . Thus *any* design asymptotically -- with enough ratings -- recovers the true ranking of sellers.

Expanded PaperInterface specification

```
∀ {n m : ℕ} [DecidableEq (GJ18InformativeRatingSystems.finiteChainOrderedPair n)]
(M : EconCSLib.Probability.FiniteRatingLDPModel (Fin n) (Fin (m + 1))) (sampleRate : Fin n → ℝ),
GJ18InformativeRatingSystems.FiniteOrdinalSourceModel M sampleRate →
  (∀ (theta : Fin n), 0 < sampleRate theta) →
    1 ≤ m →
      ∀ (p : GJ18InformativeRatingSystems.finiteChainOrderedPair n),
        Filter.Tendsto
          (fun k =>
            GJ18InformativeRatingSystems.sourceStatePairwisePk
              (GJ18InformativeRatingSystems.iidFloorSourceStateLaw M sampleRate k)
              (GJ18InformativeRatingSystems.iidFloorSourceStateScore M sampleRate k) p)
          Filter.atTop (nhds 1)
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_pairwise_objective_tendsto_one_iid_completion

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 17

Source locator: cited publication, lines 744-745

Verbatim source input

No single choice of Y and ϕ can simultaneously optimize W_k for all k : some designs may be effective in separating the best sellers from the worst quickly, but then never separate all sellers. Further, as long as $\phi(y)$ is strictly increasing, then because $R(\theta, y)$ is strictly increasing in θ , we have, for all $\theta_1 \neq \theta_2$, and all choices of Y and ϕ :

$$\lim_{k \rightarrow \infty} P_k(\theta_1, \theta_2) = 1.$$

Using the bounded convergence theorem we conclude that $\lim_{k \rightarrow \infty} W_k = 1$, independent of the design choice Y and ϕ . Thus any design asymptotically -- with enough ratings -- recovers the true ranking of sellers.

Expanded PaperInterface specification

```
∀ {n m : ℕ} [DecidableEq (GJ18InformativeRatingSystems.finiteChainOrderedPair n)]
  [Nonempty (GJ18InformativeRatingSystems.finiteChainAdjacentIndex n)]
  (M : EconCSLib.Probability.FiniteRatingLDPModel (Fin n) (Fin (m + 1))) (sampleRate : Fin n → ℝ),
  GJ18InformativeRatingSystems.FiniteOrdinalSourceModel M sampleRate →
  (∀ (theta : Fin n), 0 < sampleRate theta) →
  1 ≤ m →
  Filter.Tendsto
    (fun k =>
      GJ18InformativeRatingSystems.sourceStateWk
        (GJ18InformativeRatingSystems.iidFloorSourceStateLaw M sampleRate k)
        (GJ18InformativeRatingSystems.iidFloorSourceStateScore M sampleRate k))
    Filter.atTop (nhds 1)
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.source_uniform_ranking_objective_tendsto_one_iid_completion

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches source input

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation

Review row 18

Source locator: cited publication, lines 754-764

Verbatim source input

```
\begin{restatable}{theorem}{lemhatW}
  \label{lem:hatW}
  \begin{equation}
    \label{eq:rw}
    r \triangleq - \lim_{k \rightarrow \infty} \frac{1}{k} \log (1 - \{W\}_k) = \min_{0 \leq i < M} \inf_{a \in \mathbb{R}} \left\{ \right.
    \rightarrow g(\theta_{i+1}) I(a|\theta_{i+1}) + g(\theta_i) I(a|\theta_i) \left. \right\}
  \end{equation}
  where
   $I(a|\theta) = \sup_z \{ z a - \Lambda(z|\theta) \}$ , and  $\Lambda(z|\theta) = \log \sum_{y \in Y} \{\rho(\theta, y | Y)\}$ 
   $\rightarrow \exp(z \phi(y))$  is the log moment generating function of a single rating given to seller of type  $\theta$ .
%
%\end{restatable}
```

Approved corrected review target The archival source above is not asserted equivalent to this replacement.

For the author-approved corrected finite ordinal IID model with at least two uniformly distributed seller types, positive
↪ at-most-one-per-period match rates, a nontrivial ordinal rating scale, and valid adjacent pairs $0 \leq i \leq M-2$, one minus the
↪ state-level W_k objective has a finite real exponential rate equal to the finite representative of the support-safe
↪ adjacent-pair rate.

Archival source anchor: cited publication, lines 754-764

Recorded basis Author-approved corrected finite-chain model recorded in docs/GOVERNING_CORRECTED_MODEL_2026-07-24.md.

Expanded PaperInterface specification

```
∀ {n m : ℕ} [DecidableEq (GJ18InformativeRatingSystems.finiteChainOrderedPair n)]
  {M : EconCSLib.Probability.FiniteRatingLDPModel (Fin n) (Fin (m + 1))} {sampleRate : Fin n → ℝ}
  (model : GJ18InformativeRatingSystems.ClarifiedSourceModel M sampleRate),
  ∃ rate,
  model.correctedTheorem1Rate = ↑rate ∧
  EconCSLib.Probability.HasExponentialRate
    (fun k =>
      1 -
      GJ18InformativeRatingSystems.sourceStateWk (model.stateLaw k)
      (GJ18InformativeRatingSystems.iidFloorSourceStateScore M sampleRate k))
  rate
```

Lean proof endpoint (built separately)

GJ18InformativeRatingSystems.PaperInterface.author_approved_corrected_model_theorem1_finite_real_rate

Recorded source-to-Spec screening Verdict: not recorded

Reason: not recorded

Reviewer check: ☐ Matches approved corrected target

If left unchecked, explain the discrepancy or uncertainty below.

Reviewer annotation